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August 29, 2026

Editor
The Journal of Chemical Physics

Dear Editor,

Please find enclosed the manuscript “Algorithmic Construction and Exploitation of Topology-Constrained Refinement Models for Semiexperimental Equilibrium Structures,” submitted as an Article to *The Journal of Chemical Physics*.

The manuscript introduces MORPHEUS, a framework that turns semiexperimental refinement-model formulation into a reproducible topology-, symmetry-, and rank-audited procedure. Starting from non-redundant SONIC coordinate contracts, MORPHEUS generates the supported active space, applies constraints, predicates, and parameter classes consistently, and propagates uncertainties to chemically interpretable structural parameters.

The work addresses a general problem at the interface of rotational spectroscopy, molecular structure determination, and quantum chemistry: how to combine precise experimental data, computed vibrational and electronic corrections, topology, symmetry, and external reference information in a controlled inverse model. Validation spans acyclic, cyclic, planar, fused, bridged, and flexible molecules. A 49-atom testosterone case further shows that accurate equilibrium geometries, accelerated vibrational corrections, and generated class-constrained refinement models can be chained when isotope-rich semiexperimental analyses are not practical.

The source, runnable examples, benchmark outputs, and final-validation data are distributed with the public MORPHEUS 0.1.0rc8 release. The Supporting Information provides the numerical diagnostics and audit trails. The companion SMITH/SONIC coordinate methodology is available as arXiv:2607.16550.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I confirm that there are no conflicts of interest to declare.

Sincerely,

Vincenzo Barone